



Autonomic Nervous System

Pharmacology-Introduction

Sympathetic and parasympathetic pathways

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Scope and learning outcomes

The ANS adjusts visceral function to maintain homeostasis

By the end, students can...

- Describe ANS divisions and their anatomy.
- Explain transmitters and receptor families.
- Predict responses from receptor and organ.
- Classify drugs and apply them clinically.

The physiological scope

- Automatic CNS control of heart, vessels and bronchi.
- Also regulates gut, glands, pupil, temperature and bladder.
- Rapid visceral adjustments support homeostasis.

Somatic and autonomic motor output

An autonomic ganglion relays and distributes the signal

Somatic

- One motor neuron runs from CNS to skeletal muscle.
- Nm receptors at the neuromuscular junction; usually voluntary.

Autonomic

- Preganglionic neuron → ganglion → postganglionic neuron → effector.
- Effector is cardiac muscle, smooth muscle or a gland.
- Ganglia permit amplification, divergence and local modulation.
- Mostly involuntary; reflexes and higher CNS centers integrate output.

Motor pathway

CNS → preganglionic axon → Nn ganglion → postganglionic axon → organ



Sympathetic and parasympathetic anatomy

Origin, ganglia and fiber length predict the route

Sympathetic • thoracolumbar

- T1–L2 spinal origin; paravertebral chain or prevertebral ganglia.
- Short preganglionic, long postganglionic fibers.
- Mobilizes energy and supports fight-or-flight responses.

Parasympathetic • craniosacral

- CN III, VII, IX, X and S2–S4 outflow.
- Ganglia near or within target organ; long pre-, short postganglionic fibers.
- Supports digestion, secretion, micturition and reproductive functions.



Enteric system and adrenal medulla

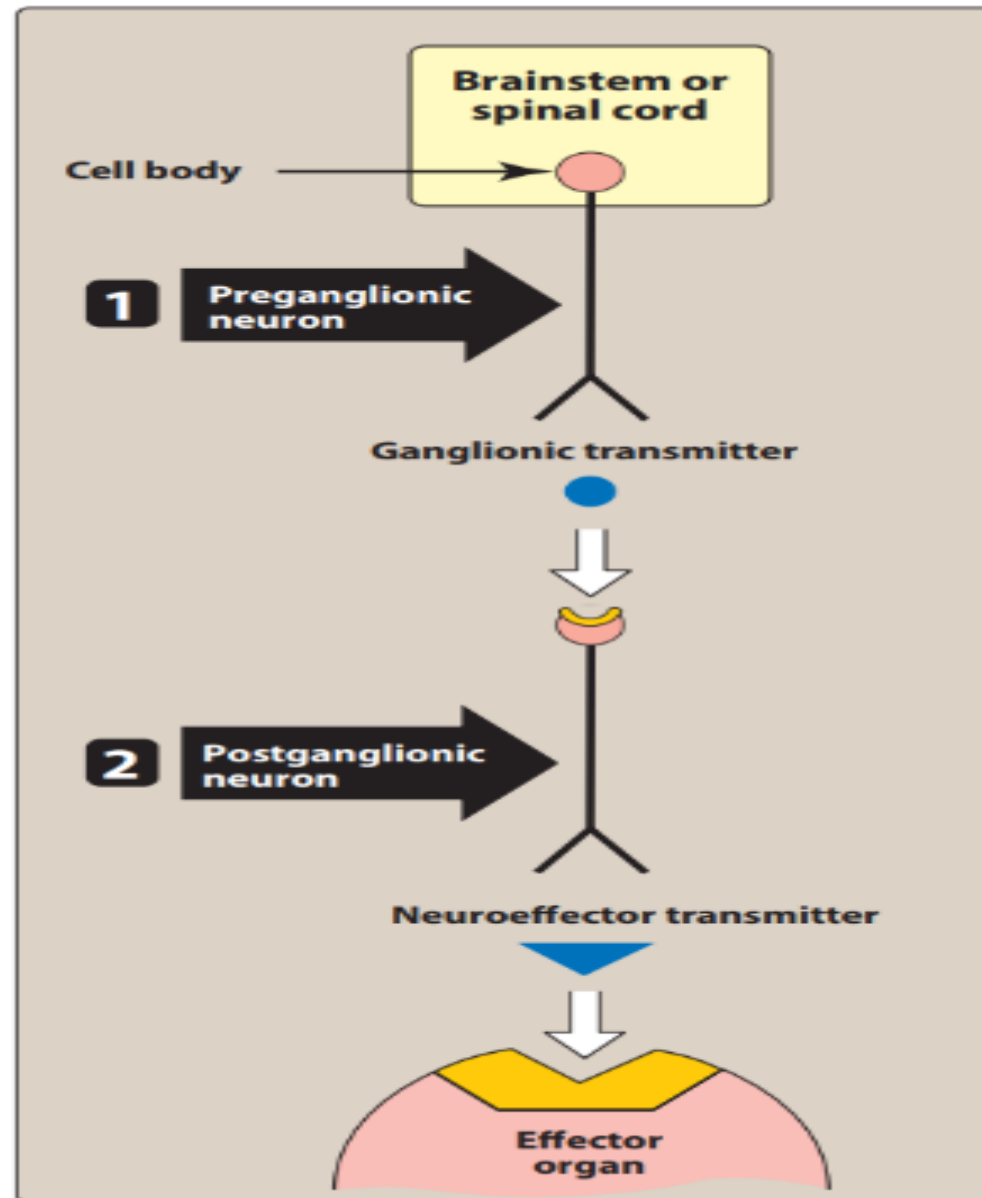
Local gut circuits and a hormonal sympathetic extension

Enteric nervous system

- Myenteric and submucosal plexuses lie in the gut wall.
- Local reflexes govern motility, secretion and intestinal blood flow.
- Autonomic input modulates the circuits.

Adrenal medulla

- Preganglionic sympathetic ACh acts on chromaffin cells.
- It functions as a modified ganglion without a postganglionic axon.
- Mainly **epinephrine** and some **norepinephrine** enter the blood.





Transmitters along the two pathways

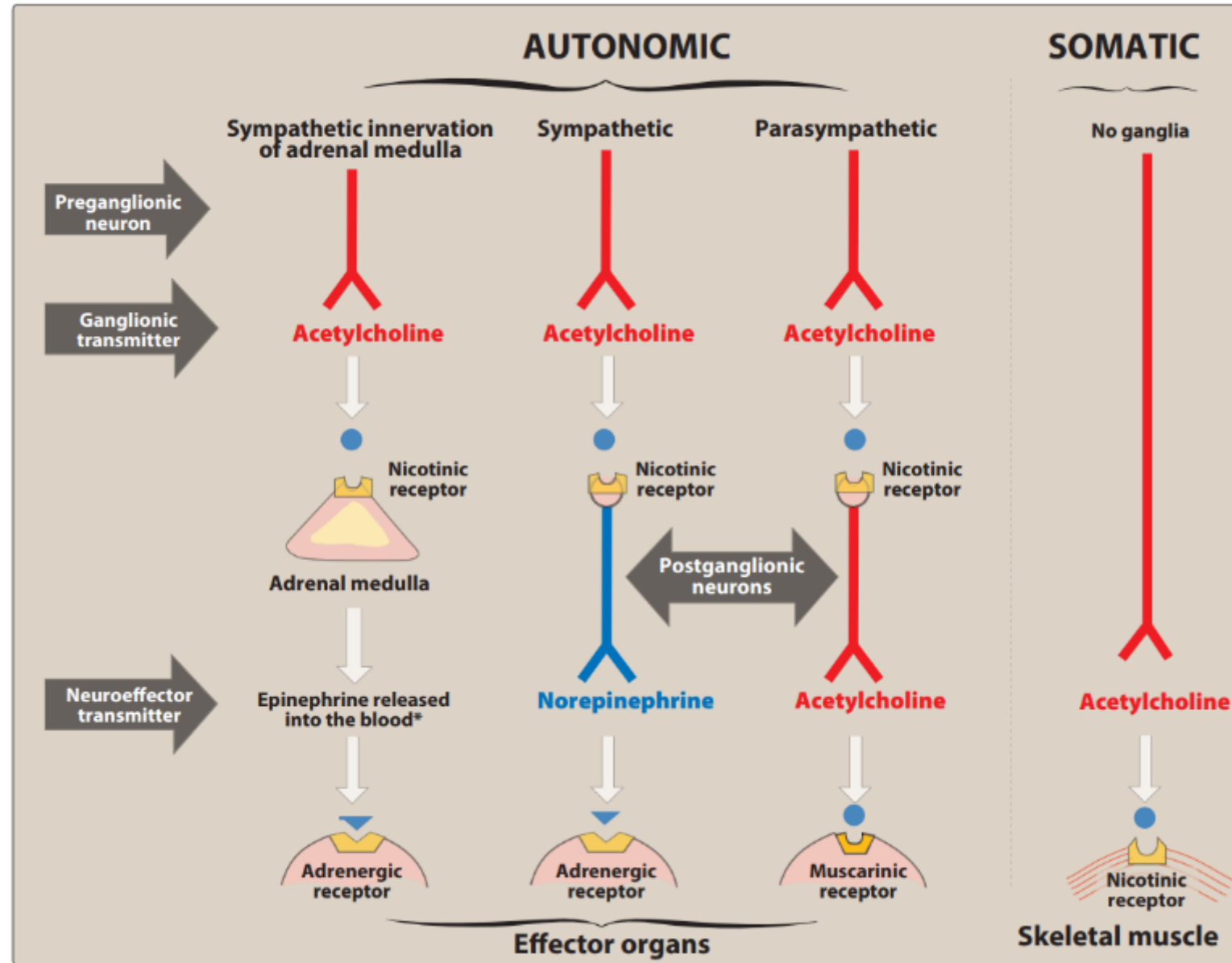
Both divisions begin with ACh at ganglionic Nn receptors

Sympathetic

- CNS → ACh → Nn ganglion → usually NE → α/β effector.
- Adrenal chromaffin cells release **epinephrine** and NE to blood.
- Other modulators: dopamine, VIP, nitric oxide and neuropeptide Y.

Parasympathetic

- CNS → ACh → Nn ganglion → ACh → muscarinic effector.
- Most postganglionic parasympathetic neurons release ACh.
- Transmitter identity and target receptor must be considered together.





Exceptions to the usual sympathetic map

Anatomical origin does not always identify the effector transmitter

- Sweat glands: sympathetic postganglionic ACh activates muscarinic receptors.
- Adrenal medulla: preganglionic ACh activates Nn on chromaffin cells; catecholamines enter blood.
- Dopamine receptors participate in renal vascular pharmacology.
- Most sympathetic effectors receive NE, but the exceptions matter clinically.



Receptor families and signaling

Fast ion channels and slower GPCR pathways

Cholinergic

- Nicotinic Nn/Nm: ligand-gated ion channels with fast depolarization.
- Muscarinic M1–M5: GPCRs for ACh at many effector sites.

Adrenergic and other GPCRs

- α and β receptors respond to NE and ***epinephrine***.
- Dopaminergic receptors matter in vascular and CNS pharmacology.
- GPCR second messengers amplify responses over seconds to minutes.



Nicotinic receptors: Nn and Nm

The subtype indicates the site and function

Nn • neuronal

- Autonomic ganglia in both sympathetic and parasympathetic routes.
- Adrenal medulla chromaffin cells.
- ACh causes rapid ganglionic transmission.

Nm • muscle

- Somatic neuromuscular junction.
- ACh triggers skeletal muscle contraction.
- Both Nn and Nm are ligand-gated ion channels.

Muscarinic receptor subtypes

M2 slows the heart; M3 contracts smooth muscle or stimulates glands

- M1: neuronal and gastric sites; supports gastric acid secretion.
- M2: heart; slows heart rate and AV conduction. **Atropine** blocks this action.
- M3: gland secretion and smooth muscle contraction, including bronchi and detrusor.
- Muscarinic receptors are GPCRs; organ location defines the response.

Alpha adrenergic receptors

$\alpha 1$ acts mainly after release; $\alpha 2$ limits transmitter release

$\alpha 1$

- Vascular smooth muscle: vasoconstriction.
- Radial iris: mydriasis; bladder neck: outlet contraction.
- **Phenylephrine** activates $\alpha 1$; **prazosin** and **tamsulosin** block $\alpha 1$.

$\alpha 2$

- Presynaptic terminals: reduced NE release.
- Central sites: reduced sympathetic outflow.
- $\alpha 1$ blockade can lower BP or relax the prostate and bladder outlet.



Beta adrenergic receptors

The receptor number predicts the dominant target

- β_1 • heart and juxtaglomerular cells: $\uparrow$ HR, contractility and renin; **dobutamine** activates, **metoprolol** blocks.
- β_2 • bronchi, uterus and skeletal muscle vessels: relaxation, bronchodilation and vasodilation; **salbutamol** activates.
- β_3 • adipose tissue and bladder detrusor: lipolysis and detrusor relaxation; **mirabegron** activates.

FUNCTIONAL ROLES & CLINICAL SIGNIFICANCE

BLADDER STORAGE PHASE (Sympathetic Dominance)

Detrusor Relaxation (β_3)

Bladder Neck Contraction (α_1A)



CLINICAL APPLICATIONS



CLINICAL APPLICATIONS

α_1 -BLOCKERS (e.g., Tamsulosin)

Blocked
 α_1A

Neck
Relaxation

Improves
Urine Flow
in BPH

α_1 -AGONISTS

Activated
 α_1A

Neck
Contraction

Treats Stress
Incontinence



Organ response: receptor + tissue

A transmitter can produce different effects in different organs

Functional examples

- NE + α_1 + arteriole $\rightarrow$ vasoconstriction.
- **Epinephrine** + β_2 + bronchus $\rightarrow$ relaxation.
- ACh + M2 + heart $\rightarrow$ slowing.

Cardiovascular effects

- β_1 raises heart rate, contractility and conduction.
- α_1 increases peripheral resistance through vasoconstriction.
- M2 slows sinus rate and AV conduction.
- Direct parasympathetic control of most peripheral vessels is limited.



Airway receptors and clinical drugs

β 2 and M3 actions oppose each other in bronchial smooth muscle

- β 2 activation relaxes bronchi and improves airflow; **salbutamol/albuterol** and **salmeterol** are β 2 agonists.
- M3 activation constricts bronchi and raises secretion; inhaled **ipratropium** or **tiotropium** blocks vagal effects.
- β 2 agonists treat reversible bronchospasm in asthma and COPD; antimuscarinics are important in COPD.
- Nonselective β blockade can oppose β 2 bronchodilation and worsen bronchospasm.



Gastrointestinal autonomic effects

Enteric circuits generate local activity under ANS modulation

- Parasympathetic activity promotes peristalsis, secretion and digestion.
- Sympathetic activity generally reduces motility, secretion and intestinal blood flow.
- Sympathetic stimulation contracts many gastrointestinal sphincters.
- Myenteric and submucosal reflexes can operate locally and receive SNS/PNS modulation.



Autonomic control of the eye

Radial and circular iris muscles produce opposing pupil responses

- α_1 contracts radial iris muscle $\rightarrow$ mydriasis; ***phenylephrine*** can dilate the pupil.
- M3 contracts sphincter pupillae $\rightarrow$ miosis; ***pilocarpine*** is a muscarinic agonist.
- M3 contracts ciliary muscle $\rightarrow$ accommodation for near vision.
- Ocular ANS drugs matter in examination and glaucoma; muscarinic agonists can improve aqueous outflow.

Skin, sweat and temperature

Sweating is sympathetic in anatomy but cholinergic at the gland

- Cutaneous vasoconstriction helps conserve heat during cold exposure.
- Sympathetic cholinergic ACh stimulates muscarinic sweat-gland receptors.
- **Atropine** may reduce sweating and impair heat loss.
- α -mediated sympathetic activity causes piloerection.

Bladder storage and emptying

Different receptors act on the detrusor and outlet

- M3 contracts detrusor during micturition; ***bethanechol*** may help selected urinary retention.
- β_3 relaxes detrusor during storage; ***mirabegron*** can reduce overactive bladder urgency.
- α_1 contracts bladder neck/prostate; ***tamsulosin*** relaxes the outlet in BPH.
- Antimuscarinic therapy such as ***oxybutynin*** can reduce detrusor overactivity.



Metabolic effects of sympathetic activation

Catecholamine signals mobilize fuel during acute demand

- Sympathetic activation increases glycogenolysis and gluconeogenesis.
- Lipolysis releases free fatty acids from adipose tissue.
- Circulating glucose rises to support acute energy use.
- β_2 stimulation also contributes to potassium movement into cells.

Baroreceptor reflex

The brainstem adjusts sympathetic and vagal output when BP changes

- Higher BP stretches carotid sinus and aortic arch baroreceptors.
- CN IX and X afferents signal the medullary NTS; afferent firing rises.
- Brainstem response: ↓ sympathetic outflow and ↑ vagal outflow.
- Heart rate, contractility and vascular tone fall; impairment may cause labile BP or orthostatic hypotension.

↑ **BP** → ↑ **baroreceptor firing** → ↓ **SNS** + ↑ **vagal output** → ↓ **HR and vascular tone**



Major classes of ANS drugs

Drugs act at receptors or alter transmitter handling

- Sympathomimetics activate adrenergic receptors or increase adrenergic transmission.
- Sympatholytics block adrenergic receptors or reduce sympathetic tone.
- Parasympathomimetics activate muscarinic receptors or increase ACh.
- Parasympatholytics block muscarinic receptors.
- Other agents alter synthesis, storage, release, reuptake or metabolism of transmitter.



Sympathomimetics

Receptor selectivity explains both effects and uses

- ***Epinephrine***: α/β agonist; used in anaphylaxis and cardiac arrest.
- ***Norepinephrine***: strong α_1 with β_1 activity; vasopressor in shock.
- ***Isoproterenol***: nonselective β agonist; raises HR and bronchodilates.
- ***Albuterol/salbutamol*** and ***salmeterol***: β_2 agonist bronchodilators in asthma/COPD.



Sympatholytics

Blocking α_1 or β receptors produces different clinical effects

- α_1 blockers cause vasodilation and relax bladder neck/prostate smooth muscle.
- **Prazosin** is an α_1 blocker; related agents such as **tamsulosin** treat BPH symptoms.
- β blockers lower HR, contractility and renin release; **metoprolol** favors β_1 , while **propranolol** blocks β_1/β_2 .
- Nonselective β blockade can worsen bronchospasm by blocking airway β_2 receptors.



Parasympathomimetics

Direct agonists and AChE inhibitors increase cholinergic effects

- Direct muscarinic agonists mimic ACh at effector receptors.
- ***Bethanechol*** stimulates bladder and GI smooth muscle.
- ***Pilocarpine*** acts on muscarinic receptors; used for xerostomia and in selected glaucoma care.
- Acetylcholinesterase inhibitors raise ACh at cholinergic synapses.



Antimuscarinic drugs

Predict therapeutic actions and adverse effects from M blockade

- **Atropine** blocks M receptors, raises HR by cardiac M2 blockade and reduces secretions.
- Inhaled **ipratropium/tiotropium** block airway muscarinic receptors and bronchodilate.
- Dry mouth, blurred vision, urinary retention and tachycardia reflect receptor blockade.
- Excessive blockade causes antimuscarinic toxicity; reduced sweating may impair heat loss.



Drugs affecting transmitter handling

Autonomic effects can arise before a receptor is reached

- Catecholamine storage, metabolism and reuptake alter transmitter availability.
- **Reserpine** depletes NE stores and is now rarely used.
- MAO inhibitors change catecholamine metabolism.
- Tricyclic antidepressants affect catecholamine reuptake and can cause autonomic adverse effects.



Clinical application: anaphylaxis

Epinephrine addresses airway and circulation through several receptors

- Pattern: hypotension, bronchospasm, urticaria and possible airway edema.
- Give IM **epinephrine** promptly for airway, breathing or circulation compromise.
- α_1 raises vascular tone and reduces mucosal edema; β_1 supports cardiac output.
- β_2 bronchodilates and reduces mediator release; reassess and follow emergency care guidance.

Pathway summary and take-home messages

Link transmitter, receptor and organ to predict the effect

Four output maps

- Sympathetic: pre ACh $\rightarrow$ Nn; usually post NE $\rightarrow$ α/β .
- Parasympathetic: pre ACh $\rightarrow$ Nn; post ACh $\rightarrow$ M.
- Adrenal medulla: ACh $\rightarrow$ Nn; **epinephrine**/NE into blood.
- Sweat glands: ACh $\rightarrow$ Nn; post ACh $\rightarrow$ M.

Remember

- Both divisions use ACh at ganglionic Nn receptors.
- The effector receptor and tissue predict organ response.
- The same map predicts drug benefit and adverse effects.

The End